

Question 1 - Problem Statement - Now that you've identified the high-level challenge, try to dive deep into it. Do a little research and come up with the specific problems under your selected challenge. For example: City women feel unsafe going out alone at night. Or Dangerous climate change is not just something that might happen in the future. It is visibly happening now and it is rapidly getting worse. Emissions are not being curbed, or waste is being disposed of directly into rivers, etc.

Problem Statement

Blind and visually impaired students face significant barriers in acquiring literacy, conceptual understanding, and creative expression due to limitations in current learning tools. Traditional Braille books often lose texture over time, making them harder to read and reducing the tactile feedback essential for building muscle memory. High-quality tactile Braille books are expensive and scarce, limiting access for each student.

Younger learners struggle to grasp Braille code due to its abstract nature and lack of engaging, multisensory teaching methods. While devices like NAB's D11 Daisy recorder allow students to capture classroom lectures, they do not support real-time, hands-on participation, causing delays in comprehension. Similarly, audio-based tools, computers, and talk-back features are hindered by challenges such as fast-paced speech, limited spelling reinforcement, and the high cognitive load required to follow along.

Moreover, blind students are often unable to draw or physically conceptualize images, impairing their understanding of diagrams, maps, and shapes, skills critical for subjects like science and mathematics. This gap widens significantly after Class 5, when subjects diversify and require more detailed and extensive learning. Without access to effective tactile, image-inclusive, and interactive educational tools, these students are left at a disadvantage, both academically and creatively.

Question 2 -

User Personas are fictional characters, which you create based on your research in order to represent the different user types. Personas can be either Contributors or Target Audiences. For example: Women Safety: Women, residing in an urban setting, age group 20-35. Climate change: Car Drivers who do not carpool, factories who do not treat their waste intern polluting the environment, residential communities who burn their garbage, etc.

User Personas – Blind Student Segments

Persona 1: Early Learner

Name: Aarav Mehta

Age: 8 years (Class 2)

Location: Small town in India

User Type: Target Audience – Early Braille User

Background:

Aarav has been blind since birth and recently mastered basic Braille. He uses the NAB D11 Daisy recorder in class to keep up with lessons but struggles to match the teacher's pace. He learns better with hands-on activities but has little access to tactile diagrams or textured illustrations.

Frustrations:

- Audio lessons are too fast and hard to follow
- No access to affordable tactile books with both Braille and images
- Finds spelling and word recognition difficult without visual reference

Goals & Needs:

- Reinforce Braille literacy through fun, interactive methods
 - Build muscle memory for letters, words, and shapes
 - Understand simple diagrams and objects through touch
 - Keep up with teacher's pace without losing focus
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Persona 2: Transitional Learner

Name: Fatima Khan

Age: 11 years (Class 6)

Location: City school with special education section

User Type: Target Audience – Mid-level Student Adjusting to Syllabus Change

Background:

Fatima is confident in reading and writing Braille but struggles after the syllabus expansion post-Class 5. She now needs to learn complex science, geography, and math concepts with detailed

diagrams and charts. Current Braille resources don't provide enough tactile representation, making abstract topics harder to grasp.

Frustrations:

- Shortage of tactile diagrams and models for higher-level subjects
- Overwhelmed by large volumes of information in Braille format
- Delays in learning due to dependence on audio playback

Goals & Needs:

- Interpret diagrams, maps, and graphs in a tactile format
 - Manage and organize information for multiple subjects
 - Practice drawing and tactile sketching for better conceptualization
 - Learn new vocabulary and spellings specific to each subject
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Persona 3: Exam-Oriented Learner

Name: Rohan Sharma

Age: 15 years (Class 10)

Location: Urban high school with inclusive classrooms

User Type: Target Audience – Senior Student Preparing for Board Exams

Background:

Rohan is an independent Braille reader with strong academic ambition. He faces time pressure due to upcoming board exams. Complex diagrams in biology, geography, and mathematics are a major challenge without tactile representation. He often relies on friends or teachers for verbal explanations, which slows revision.

Frustrations:

- Difficulty revising large portions of syllabus quickly
- Lack of affordable high-quality tactile study material
- Inability to independently “see” and practice diagram-based questions

Goals & Needs:

- Access to tactile diagrams and models for quick, independent study
- Efficient note-taking and organization for exam preparation
- Revision tools that combine Braille text with tactile imagery
- Minimize reliance on others for learning and diagram interpretation

Describe your product / service: Use a story to depict how your user's life will be impacted! Please add a flowchart or an articulate representation of the journey the user takes while interacting with your solution.

Product / Service Description – Storytelling Format

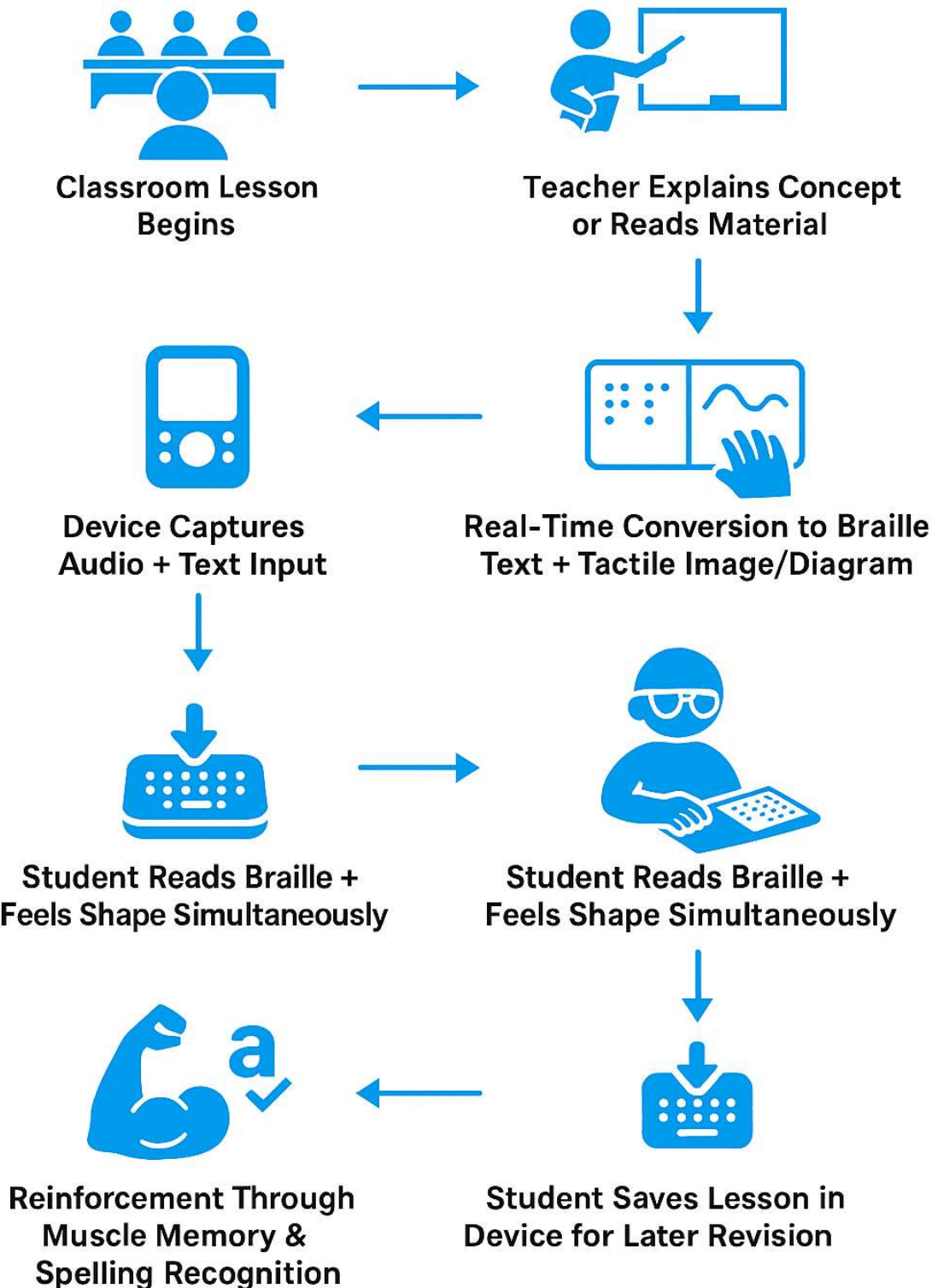
Aarav, an 8-year-old in Class 3, starts his day in school with his NAB Daisy recorder beside him. The teacher introduces a new topic on “Animals and Their Habitats.” Aarav presses record as usual, but this time, instead of passively listening later, he reaches into his bag and pulls out your **Portable Braille-and-Tactile Learning Device**.

The device syncs with the teacher’s lesson in real time. As the topic is read aloud, Aarav feels raised Braille text forming on the device’s tactile pad, alongside textured illustrations of animals and their environments. His fingers trace the shape of a fish while reading the word “fish” in Braille — reinforcing spelling, meaning, and shape in a single moment.

Fatima, a Class 6 student, uses the same device in her geography class. When the teacher explains “river systems,” the device presents a raised tactile map with labeled Braille points, helping her instantly understand the flow from mountains to oceans without waiting for a Braille book to arrive weeks later.

Rohan, in Class 10, uses it for quick revision before his board exams. He uploads past paper diagrams, which the device instantly converts into tactile graphics with labeled Braille. This means he can independently study and revise diagrams in biology or geography without relying on others to describe them.

From early learners to exam-bound students, the device transforms the way they learn — moving from **delayed, audio-heavy, and resource-scarce education** to **real-time, multisensory, self-paced learning**.



**Better Understanding ➡ Improved Confidence
➡ Higher Academic Performance**

How the startups innovation can make a social impact in the society?

Social Impact of the Portable Braille-and-Tactile Learning Device

1. Inclusive Education for All

- Provides visually impaired students with **equal access to real-time learning**, breaking the dependency on delayed Braille books or verbal explanations.
- Encourages **participation in class**, rather than passive listening, fostering active engagement.

2. Bridging the Learning Gap

- Supports early learners in **mastering Braille faster** through combined text-and-image tactile feedback.
- Helps students transitioning to higher classes **understand complex diagrams and abstract concepts** that are otherwise inaccessible.
- Prepares senior students for exams **independently**, without relying on peers or teachers.

3. Empowering Independence

- Gives blind students the **tools to learn, revise, and explore** without constant assistance.
- Builds **muscle memory and spelling accuracy**, enabling better literacy outcomes.

4. Affordability & Accessibility

- Reduces reliance on costly, scarce tactile Braille books.
- Offers a **portable, scalable solution** that can be distributed in rural and urban schools alike.

5. Enhancing Employment Prospects

- Early and effective education opens pathways to **higher education and skilled jobs** for visually impaired students.
- Promotes **digital literacy**, preparing them for modern work environments.

6. Community & Policy Impact

- Demonstrates that **inclusive design** benefits marginalized groups, influencing policy and educational reforms.
- Strengthens advocacy for **accessible learning technology** in public education systems.

SOCIAL IMPACT OF THE PORTABLE BRAILLE-AND- TACTILE LEARNING DEVICE

ENHANCING EMPLOYMENT PROSPECTS

Expanding pathways
to higher education

AFFORDABILITY & ACCESSIBILITY

Reducing reliance on costly
books. Portable & scalable

EMPOWERING INDEPENDENCE

Build muscle memory, spelling accuracy
Foster self-directed learning

BRIDGING THE LEARNING GAP

Help early learners master Braille, aiding subject transitions

INCLUSIVE EDUCATION FOR ALL

Providing equal access to real-time learning via-rasing